

Data from long-term experiments in temperate croplands to evaluate soil organic carbon models

Fujisaki, Kenji; Ferchaud, Fabien; Clivot, Hugues; Bruni, Elisa; Guenet, Bertrand; Pichot, Christian; Versini, Antoine; Baudin, François; Bispo, Antonio; Peylin, Philippe; Martin, Manuel P.; Jensen, Johannes L.; Eriksen, Jørgen; Chenu, Claire; Gregory, Andrew S.; Glendining, Margaret J.; Merbach, Ines; Beaudoin, Nicolas; Mary, Bruno; Mollier, Alain; Tison, Gilles; Montagnier, Christophe; Chabbi, Abad; Vertès, Françoise; Cadéro, Alice; Graux, Anne-Isabelle; Pellerin, Sylvain; Levavasasseur, Florent; Gilles, Manon; Morvan, Thierry; Resseguier, Camille; Milesi, Luis; Irizar, Alicia; Andriulo, Adrià; Mistou, Marie-Noël; Butier, Arnaud; Bertrand, Michel; Autret, Bénédicte; Jeuffroy, Marie-Hélène; Grandeau, Gilles; Doré, Thierry; Cellier, Vincent; Berthier, Alain; Darras, Sébastien; Audebert, Guillaume; Pasquier, Ludovic; Ecalte, Fabien; Savoie, Antoine; Schiedung, Marcus; Poeplau, Christopher; Maaroufi, Nadia I.; Kätterer, Thomas; Bolinder, Martin A.; Sanderman, Jonathan; Barré, Pierre (2026.0)

ABSTRACT ORIGINAL

Soil organic carbon (SOC) models need independent evaluation against field measurements, but those latter are rarely publicly available and harmonized. In this study, we collected and shared data from 167 agronomic treatments in 34 agronomic long-term experiments (LTEs) located in temperate croplands, allowing the evaluation of several soil organic C models such as RothC, Century, AMG, MIMICS, ICBM, Millennial, and CTOOL. The dataset includes climate data, soil properties, C inputs from crops (n = 4588 records) and organic amendments, irrigation data, monthly soil cover, as well as SOC stock measurements in the topsoil layer (n = 1328 records). Climate, soil moisture, and soil temperature data were extracted from daily climate databases. Carbon inputs from crops were calculated from observed yields and harvest index, with some harvest index values estimated, combined with crop allometric coefficients from the literature. Descriptions of LTE, agronomic treatments, methodological metadata, and a part of the code, accompanies the dataset. The dataset can be reused to evaluate single SOC models, or to evaluate an ensemble of models. © The Author(s) 2026.